

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df=pd.read_csv('USA_Housing.csv')
```

```
df.head()
```

	Avg. Area Income	Avg. Area House Age	Avg. Area Number of Rooms	\
0	79545.458574	5.682861	7.009188	
1	79248.642455	6.002900	6.730821	
2	61287.067179	5.865890	8.512727	
3	63345.240046	7.188236	5.586729	
4	59982.197226	5.040555	7.839388	

	Avg. Area Number of Bedrooms	Area Population	Price	\
0	4.09	23086.800503	1.059034e+06	
1	3.09	40173.072174	1.505891e+06	
2	5.13	36882.159400	1.058988e+06	
3	3.26	34310.242831	1.260617e+06	
4	4.23	26354.109472	6.309435e+05	

	Address
0	208 Michael Ferry Apt. 674\nLaurabury, NE 3701...
1	188 Johnson Views Suite 079\nLake Kathleen, CA...
2	9127 Elizabeth Stravenue\nDanielstown, WI 06482...
3	USS Barnett\nFP0 AP 44820
4	USNS Raymond\nFP0 AE 09386

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 5000 entries, 0 to 4999
```

```
Data columns (total 7 columns):
```

#	Column	Non-Null Count	Dtype
0	Avg. Area Income	5000 non-null	float64
1	Avg. Area House Age	5000 non-null	float64
2	Avg. Area Number of Rooms	5000 non-null	float64
3	Avg. Area Number of Bedrooms	5000 non-null	float64
4	Area Population	5000 non-null	float64
5	Price	5000 non-null	float64
6	Address	5000 non-null	object

```
dtypes: float64(6), object(1)
```

```
memory usage: 273.6+ KB
```

```
y=df['Price']
```

```
y
```

```

0      1.059034e+06
1      1.505891e+06
2      1.058988e+06
3      1.260617e+06
4      6.309435e+05

```

```

...
4995    1.060194e+06
4996    1.482618e+06
4997    1.030730e+06
4998    1.198657e+06
4999    1.298950e+06

```

Name: Price, Length: 5000, dtype: float64

```
X=df[['Avg. Area Income','Avg. Area House Age','Avg. Area Number of
Rooms','Avg. Area Number of Bedrooms' , 'Area Population']]
```

X

	Avg. Area Income	Avg. Area House Age	Avg. Area Number of Rooms
0	79545.458574	5.682861	7.009188
1	79248.642455	6.002900	6.730821
2	61287.067179	5.865890	8.512727
3	63345.240046	7.188236	5.586729
4	59982.197226	5.040555	7.839388
...	...	...	...
4995	60567.944140	7.830362	6.137356
4996	78491.275435	6.999135	6.576763
4997	63390.686886	7.250591	4.805081
4998	68001.331235	5.534388	7.130144
4999	65510.581804	5.992305	6.792336

	Avg. Area Number of Bedrooms	Area Population
0	4.09	23086.800503
1	3.09	40173.072174
2	5.13	36882.159400
3	3.26	34310.242831
4	4.23	26354.109472
...	...	...
4995	3.46	22837.361035

4996	4.02	25616.115489
4997	2.13	33266.145490
4998	5.44	42625.620156
4999	4.07	46501.283803

[5000 rows x 5 columns]

`type(y)`

`pandas.core.series.Series`

`type(X)`

`pandas.core.frame.DataFrame`

`from sklearn.model_selection import train_test_split`

`X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=42)`

`X_train`

	Avg. Area Income	Avg. Area House Age	Avg. Area Number of Rooms
4227	66547.016454	5.846095	6.847298
4676	53722.008599	6.401391	7.787764
800	64838.492899	6.437157	8.699544
3671	67097.092120	6.086754	7.211963
4193	75245.465436	8.167820	7.420100
...	...	...	...
4426	76223.561256	6.371627	5.342217
466	56685.014442	6.958045	7.502115
3092	66195.337714	6.507971	6.611861
3772	58694.515017	7.394768	9.269453
860	61162.580254	5.896316	7.880521
	Avg. Area Number of Bedrooms	Area Population	
4227	4.13	27850.822901	
4676	3.30	47649.224665	
800	4.02	32921.010068	
3671	3.05	27191.506877	

4193	3.42	37410.669928
...	...	...
4426	2.42	30165.337445
466	3.38	43322.166854
3092	3.14	37288.923574
3772	4.32	49960.977236
860	6.04	36033.701431

[4000 rows x 5 columns]

X_test

	Avg. Area Income	Avg. Area House Age	Avg. Area Number of Rooms
\			
1501	61907.593345	7.017838	6.440256
2586	57160.202243	6.893260	6.921532
2653	70190.796445	6.745054	6.662567
1055	69316.796889	6.300409	7.873576
705	72991.481649	3.412866	6.494081
...	...	...	...
4711	77267.656264	3.939501	8.342808
2313	75967.135085	5.939370	6.111658
3214	81013.615294	7.149797	7.239105
2732	86762.882864	6.530193	5.106962
1926	67071.830617	4.935155	7.632398

	Avg. Area Number of Bedrooms	Area Population
1501	3.25	43828.947207
2586	3.13	43467.147035
2653	2.01	29215.136112
1055	4.28	24448.211461
705	2.48	50626.495426
...	...	...
4711	6.09	22487.712072
2313	2.32	38897.091584
3214	5.44	45472.049451
2732	2.09	47724.581355
1926	5.04	32084.743400

[1000 rows x 5 columns]

y_train

```
4227    1.094880e+06
4676    1.300389e+06
800     1.382172e+06
3671    1.027428e+06
4193    1.562887e+06
```

...

```
4426    1.023944e+06
466     1.223101e+06
3092    1.318598e+06
3772    1.708631e+06
860     1.060898e+06
```

Name: Price, Length: 4000, dtype: float64

y_test

```
1501    1.339096e+06
2586    1.251794e+06
2653    1.340095e+06
1055    1.431508e+06
705     1.042374e+06
```

...

```
4711    1.107031e+06
2313    1.405505e+06
3214    1.924156e+06
2732    1.571254e+06
1926    8.831475e+05
```

Name: Price, Length: 1000, dtype: float64

```
from sklearn.linear_model import LinearRegression
```

```
modell=LinearRegression()
```

```
modell.fit(X_train,y_train)
```

```
LinearRegression()
```

```
y_predict=modell.predict(X_test)
```

y_predict

```
array([1308587.92699714, 1237037.22949391, 1243429.34030722,
       1228900.21360438, 1063320.90710747, 1544058.05034817,
       1094774.70493034,  833284.72339248,  788412.85578756,
       1469714.86615696,  671728.43662092, 1606818.21977767,
       1004166.61331045, 1796798.97595973, 1288566.96220949,
       1087782.93301086, 1423072.3749249 , 1078178.68169646,
        802286.03537917,  930761.03695739, 1134829.86477803,
        916398.42023083, 1489972.69335353, 1284580.15538841,
       1582071.35322685, 1132519.15992002, 1089888.39644489,
```

974510.51872177, 924057.96820805, 1740759.72092213,
1286481.59512379, 1621289.95171618, 1435264.20161695,
1234014.77924529, 1485434.5730037, 1718335.00753593,
1538953.74882758, 777106.64791831, 1765201.52243597,
1175972.14199831, 1553707.94323478, 897703.67505165,
1371049.803266, 845281.72310351, 1201022.89803856,
1133285.98450789, 1363128.14557396, 1449814.0876822,
1574363.90467323, 1233577.50266003, 1484464.01606146,
1295276.58943534, 1222136.77335411, 990124.41659661,
1693824.96035776, 1823785.05665058, 1136495.63903301,
1282164.40305655, 1327292.05443206, 1353355.51909081,
966265.44346008, 661906.6391724, 1533750.56861038,
1002479.76053604, 995799.79959577, 1567349.59707181,
1500813.63482544, 1090078.00056468, 1820964.84741614,
1479856.167243, 902785.3021607, 1494542.21679532,
1378859.29762395, 962610.88478541, 712800.76347545,
1565650.37425334, 1149218.97654367, 931311.21938312,
1600923.95574322, 506875.42639308, 1592924.03164877,
1292023.54953405, 681260.98814038, 432977.16110056,
1395334.63471246, 696834.50740119, 663613.86440665,
1030075.82567097, 1485134.02140083, 1320071.75329084,
1271264.11329325, 1422274.66428959, 671601.89583304,
1149995.64271504, 1261301.82190822, 784691.75366812,
1189915.30485975, 1014887.17531276, 1405378.85715023,
1539076.97995898, 593301.15079239, 1391801.76393913,
1262742.70864065, 1875132.68141912, 2336899.33387139,
946302.25269136, 1315655.58002691, 953107.00437499,
1831758.27483121, 1625882.69024171, 1639245.35999709,
1267016.02457525, 1804840.84718802, 1220829.46104284,
850670.85934729, 1584187.59595389, 761904.58839524,
1360032.80169357, 1186062.16197455, 1559470.32105814,
1770323.74482733, 1608248.53684128, 1573661.86962713,
1531208.64376156, 1812547.75249854, 1124965.37862327,
669517.22471181, 978249.45590812, 1316294.24560539,
1627775.4894282, 1343329.23943851, 1088746.21767489,
1516813.82322275, 1333391.56256443, 1421961.53699962,
1527965.13951197, 1674048.56946842, 1301125.76637761,
843902.14276083, 1439397.74699529, 1857462.44038355,
1090388.23396924, 1845464.91767028, 1272015.37672498,
2036878.56552091, 882686.73444809, 954142.09498624,
1116676.58455452, 1395124.12307515, 1510721.91006236,
952949.66077836, 969883.09423412, 1320445.66982262,
1494548.61488366, 1311797.03936272, 764593.88740317,
1729945.04026266, 966390.95323417, 1980014.56655046,
1253548.560401, 1378043.01470128, 1465705.85673404,
1041643.25777121, 2016178.1230771, 1425286.31755915,
783766.05598717, 826322.77792554, 1753896.36530072,
1077079.69502203, 2073701.72252865, 1735967.08462983,
1049433.84155988, 1829294.89844213, 1128895.79613832,

1629029.3528393 , 1634470.42260782, 1524785.7096465 ,
722156.79424196, 740342.6213115 , 547622.83148213,
828931.14665451, 1067364.15017839, 969253.52659167,
1227292.55245908, 871755.1279939 , 504202.02965952,
1200044.47873889, 1062310.88487562, 1519500.9929722 ,
1329223.54689472, 1386542.64820107, 1320937.40536445,
1204359.90632443, 1240619.19034638, 1365626.91484043,
1769953.37847142, 1329720.91129388, 1376332.51764085,
1298990.36407913, 834744.4031073 , 1272066.20149392,
1451499.03548565, 1289190.82085851, 896873.03598678,
990437.07011067, 1357842.65851649, 1649364.37150859,
779324.24925057, 1005588.56759432, 1467831.8143511 ,
1124908.45994284, 1142768.9365308 , 1250196.34738779,
1105424.47695273, 622294.93861793, 1841657.41212501,
1182731.53995538, 609160.57633679, 1410661.53419237,
1263529.22101432, 1119673.80759963, 1145399.95692925,
500449.33291 , 1109883.63039201, 1626981.199574 ,
797675.84432516, 1407092.10326845, 1560733.66870718,
1698894.16984378, 1714320.59543043, 1216152.65562669,
1114774.40354027, 1001067.89613646, 1065269.68436401,
1625179.02290118, 1593962.78450191, 1526221.35988262,
1129983.77332897, 735887.82286918, 1593872.8090903 ,
1732256.51293184, 1264953.46131018, 1216712.64382556,
1539828.39332814, 1748273.31153947, 544482.71161728,
1177808.65915499, 1404019.93663788, 1125393.51909489,
545088.45128059, 1290749.88927717, 1658361.89913428,
1487203.28008188, 1291143.0383848 , 2050455.16760411,
768874.19747935, 1863631.84714332, 929819.17764119,
1761083.62018149, 1659001.75083626, 1247438.96208962,
713427.40837841, 1778756.53025758, 461932.4961182 ,
2167306.95873247, 664672.42431932, 1801836.62367777,
1265099.18805305, 1139217.30086043, 1173989.48300885,
1748192.41949242, 1147793.31302128, 1029923.90766774,
947126.10538068, 1182469.49434635, 927601.20005187,
1566514.61685729, 1175827.55938891, 388087.54801502,
1288959.37006079, 986658.1919734 , 1340019.64945834,
1044418.84541027, 1286342.56336201, 1531000.55785292,
1667768.88543247, 1084796.51405753, 720967.4306256 ,
811997.72885717, 1328871.66384378, 1519499.84416171,
1379415.47205369, 1457746.58240954, 998979.73972497,
2252173.82126768, 1491966.70905195, 1349013.08643529,
1042666.25727847, 486559.46844387, 1792549.96363352,
1193458.84754761, 1247203.2650348 , 1122698.47045959,
1004013.28018429, 1699121.45816191, 1504252.14207202,
1432835.58985504, 1342786.68025844, 1601414.77106583,
821506.76488296, 1126578.53904187, 1291476.44544106,
1698968.4422244 , 1232602.93325311, 1489643.67510259,
1233262.35005618, 1135048.11749495, 1732127.82573453,
1348162.21747094, 621085.06641237, 1501389.10755797,

1201749.81984471, 1326727.70232223, 1097551.72695628,
1221290.6626145 , 788262.02543974, 935562.28249739,
1287412.17867166, 1473888.67757694, 609893.86640577,
1183181.37838201, 1035484.44410104, 809285.45807012,
655567.08949461, 983743.80081262, 986631.41757991,
1320596.67974152, 1196954.70045335, 961010.91970451,
1275847.42178904, 1425242.66569497, 1654320.5462453 ,
1521722.43867584, 1396978.62698658, 1413827.92718118,
1219885.02000764, 1232374.5990155 , 1654016.63386521,
1328324.05157988, 1321091.57152418, 1299216.74514609,
1347305.90163689, 1688687.80000919, 1574519.36949703,
1045335.69889304, 1356261.70016912, 1439514.97152487,
1504156.15649894, 1138145.82597045, 1500051.72294308,
1486912.30438018, 385027.92486015, 1216217.42146564,
1726329.91434959, 1207805.98489455, 894601.19092648,
1240643.83179271, 1308440.22054425, 1437359.37891269,
893239.6816693 , 1792787.36750054, 2076933.42343334,
922567.65702038, 1458351.25849111, 1249772.50297827,
1370805.23907784, 1121245.98909414, 963231.49570935,
1327241.12918529, 1187327.23418126, 943716.62278268,
424854.57298743, 1223724.58352509, 1411756.82716417,
1236002.32151855, 1449021.08575886, 1411706.423207 ,
926881.67781004, 1006631.71873016, 1325140.65527558,
470014.10664436, 1568408.54620602, 1071690.67501273,
1261709.43785505, 983184.15258562, 1086377.76905192,
726363.60820814, 814172.12713779, 803025.5488858 ,
1619615.99437041, 459335.13663739, 1300390.9001114 ,
961516.33244628, 1025131.04020116, 1384778.89206789,
1235853.66369932, 769788.89202082, 1026166.69741528,
1405638.84930905, 1756291.1165281 , 1280487.11747006,
622591.98844858, 1449923.84156208, 1198418.94698622,
1356968.16583347, 1079271.72703246, 1352329.77296876,
1057107.45852964, 1290462.77150162, 705163.48550206,
1200364.66997161, 1507695.00401142, 962337.84598511,
1204232.84729686, 1571860.58065027, 1209464.73857427,
1549865.08072211, 769645.60725025, 1404674.43922922,
1240637.52783313, 1029500.08001662, 1335457.58612514,
1088594.03736602, 1047144.04180958, 1388984.12610535,
918924.01605126, 1096407.50174631, 1406966.35948271,
806952.29876878, 1196187.41426752, 1010141.32136431,
1394711.984787 , 1257521.42006973, 1299693.11290862,
1294970.9428869 , 1723105.21964877, 1711247.39032038,
1174489.95194168, 897740.74110047, 1356640.83884517,
1219851.03447621, 1366674.22214225, 1644472.14734927,
724989.37808751, 1570999.00692545, 924989.36290744,
673283.26497766, 1589694.70648699, 1373578.19145125,
1236758.80570297, 1939038.59441329, 1248340.03854229,
1393184.15037987, 1703469.17720753, 1059230.86749291,
1268676.36389762, 1304697.18785127, 892460.1227714 ,

964864.99058535, 818820.07760037, 1401580.5474752 ,
1180874.94215921, 1259247.43803538, 1702902.0449319 ,
1649087.11983931, 1731289.88335283, 1125719.05993057,
865172.86237642, 968232.22605858, 784711.55849911,
781822.82332277, 1209737.47415562, 972635.23672537,
1576919.8361663 , 1060076.42506716, 694196.84219498,
1610587.67992985, 1309658.6163039 , 1141463.96987352,
1205906.9859995 , 1281336.1844416 , 1171886.92407766,
1567896.4506818 , 1322659.49232192, 1166515.84227974,
789697.38830608, 1143720.96771758, 648247.70636935,
1071126.44204693, 1428574.57982083, 1424322.83234031,
1057501.45875355, 1430509.55156425, 1324079.70373571,
1292014.8973667 , 1071126.32745223, 1287175.72484535,
1019167.68338732, 1037478.15144031, 1049936.76364954,
731789.18467522, 782568.77309993, 1002203.86002329,
1067321.67597129, 1124201.97076048, 1457954.83235594,
1108354.56185648, 1598770.07901318, 1176022.85579896,
1096986.73368095, 1247383.41805669, 1363559.4817819 ,
773372.64579591, 822330.64613243, 993393.81206808,
1543915.09520911, 1493186.91902704, 1175891.67528439,
1253692.12639549, 1521005.76686495, 1627802.51180515,
1686082.39992492, 1980124.37797531, 977460.80106607,
1109687.22060379, 1113007.40549553, 1211544.64900645,
1407486.32673172, 789313.25630073, 1234898.66189976,
1209078.69432756, 1803383.9507997 , 2114888.79328687,
1118853.92157715, 1206707.38468672, 1352852.993439 ,
1396175.67664005, 1546374.75741914, 666842.59066058,
1481420.71952374, 1021477.4551858 , 968486.88335943,
1424082.26366213, 1914787.24107111, 914204.87835342,
1489860.90731096, 693922.34630688, 1031062.51066014,
1352126.0325433 , 1373512.52703602, 1049109.83383307,
1486525.47762685, 1310520.39595894, 1647941.99311814,
1283575.24055879, 1452989.07762719, 1429414.83148135,
1034083.03488008, 1134634.5146035 , 1403564.88673527,
1584896.61276133, 1746823.45559135, 1466283.25973607,
1294745.41392556, 1051682.47255767, 1101548.76032296,
1465167.00089561, 1382725.06572414, 939224.93035434,
1337350.76799284, 1120979.59497988, 1319831.0377081 ,
558962.93258742, 1401611.09978773, 1354544.07306087,
1306334.89969526, 2086259.17091894, 1904134.11165434,
539895.35619553, 1525528.05146349, 1235213.44649991,
750639.88255422, 1127965.98258833, 1471490.85051118,
619419.48787537, 1114997.61194519, 1061464.69123039,
1013604.11379752, 1581434.85227387, 1079499.50514152,
1055056.29175909, 1034589.95466434, 1224314.04495528,
887268.18862849, 1516254.20491696, 1865327.56742204,
1027371.06429478, 1490055.38180638, 1337289.95134359,
1119421.33157691, 979092.31019086, 679557.25776168,
1325905.50178092, 1171590.73226988, 1567491.12331888,

1143856.19166598, 1126148.64416527, 864154.38857817,
490147.41546395, 1291071.27278545, 1038384.50618656,
1662679.55956637, 1160624.90934354, 2037284.10126505,
1516262.89327408, 1229075.2331131 , 1547218.08935871,
1434753.6539263 , 1709132.86008207, 1281184.09975617,
1524044.07660681, 968015.40091394, 1293671.18547207,
1106474.84699341, 1431104.22750471, 1712068.30269008,
1160254.52501101, 655742.41586225, 1226710.98389822,
927585.91056611, 1267635.28046212, 891680.82011524,
1234379.98343431, 921258.67102351, 973622.38506852,
1211118.4915946 , 1152861.17190994, 1769241.04652278,
1185559.02437273, 982633.08890869, 1332532.6146074 ,
1642473.93936951, 965382.57146955, 1281975.93146394,
1406343.90581985, 677673.85740157, 1209557.69139042,
732921.10636363, 1682132.4524236 , 872620.43651325,
1136143.09175212, 510258.7550623 , 1288102.88621638,
1371033.3346398 , 1310113.78224326, 1436855.31241042,
1051232.72974669, 684828.48739361, 1101019.83617934,
1567983.51294801, 1018012.68216361, 1050698.11270312,
825827.00943885, 1173621.66011216, 1591981.75730133,
1595728.99149165, 1360891.37529113, 1192754.82897969,
655836.47938466, 1560112.7853046 , 813169.25379871,
1132145.75967511, 1564590.92451706, 1118669.76459779,
1569760.45331536, 1383018.33137042, 1609536.2300856 ,
1045549.71074357, 1174361.92171486, 1447869.79959177,
1824783.76109702, 1639844.91468839, 1488884.78527564,
1427514.92585418, 1231681.33305912, 1368014.03189635,
1424922.73584767, 1551131.49668313, 914701.22781707,
1249841.22219038, 258045.70919602, 1230968.70157337,
1126752.12639181, 858783.44109428, 739043.62099808,
1044214.90471311, 1112387.60767918, 1223250.56468727,
1447909.23403619, 1328619.03653258, 1049307.01433342,
769590.67267147, 1751850.88833512, 1463027.897591 ,
2139919.10952291, 1718412.1371305 , 1115993.45943508,
1995828.29498941, 1193379.31596138, 660188.56028408,
660678.46196101, 1798655.55037073, 470131.44658409,
1364102.16506243, 889323.70684001, 706548.24954165,
1399891.22458689, 1416306.27235891, 967240.32613214,
1729652.56536646, 869958.83238876, 1437817.09450817,
1122484.53204022, 1673255.03691676, 1440039.59230989,
1500539.40538973, 971584.27416956, 1043184.935673 ,
1317543.16932928, 1007627.13026089, 1323354.40365541,
1135667.17490178, 1510351.76679851, 993056.44789484,
1208831.6583276 , 1988655.64291084, 1230200.72390674,
1390403.72276626, 1262914.99801435, 1582327.50953688,
1028414.83031315, 1672107.67442239, 1265752.8295915 ,
1236601.43306984, 329710.11232961, 717954.1128098 ,
1103733.13974486, 1188712.89825671, 1261796.72795411,
1193848.8215225 , 1303384.58750888, 1076900.5357587 ,

1169750.68418026, 1458470.38909803, 1190650.84525978,
1360612.2033834 , 1497401.76740699, 916161.01831818,
1311284.98583137, 1579839.09403304, 1155915.5454168 ,
1425748.49376165, 1287287.54555809, 1153862.22489316,
1825980.94614764, 1049734.1522132 , 909542.47470268,
1253936.42188605, 1467206.44416842, 1639991.73405582,
1521767.77491785, 1019769.20382372, 652300.40469468,
643377.65271697, 1142025.9127117 , 1263237.98299683,
1258087.11819232, 589606.05536976, 1187007.16456487,
1121740.50402122, 1808959.71031358, 1800055.63690805,
1123524.63673938, 1425287.86106934, 1510529.71215077,
1323072.46306216, 1443101.22620898, 1296655.19033918,
1551943.68696653, 1237292.00298087, 665445.49485864,
1262709.17153413, 1048886.41492071, 1867303.54970962,
1562894.27356947, 731321.70170499, 1239100.56464228,
909236.28293377, 1752100.42286093, 1602831.78580355,
1504851.06631647, 1411568.94781492, 1394422.38818576,
370722.62126894, 1672311.97769409, 2120154.65355426,
421201.54386786, 1486648.19169931, 1200115.30672021,
1531544.96637764, 1031782.76469213, 842914.91778182,
838950.40676634, 596819.74942491, 855574.21947051,
1040351.51174867, 1499196.39432647, 1353188.18645183,
1167128.79448732, 980462.42074901, 1615652.70345418,
1192255.12726459, 1280277.40202466, 542769.58059301,
1121564.11390543, 1430520.80041271, 1187720.59708794,
1293444.79096409, 1344677.83092672, 1183418.13056097,
1508087.09800629, 1265155.24508926, 1342605.87097162,
1277584.98121803, 733241.48650893, 1553389.38956826,
1398171.05352941, 1187021.49985599, 1371024.2900689 ,
1108983.89166018, 975275.64367687, 1202021.72545496,
1933557.6627894 , 753137.60441143, 972296.00805803,
1448918.48607893, 1859400.97278726, 914399.5769572 ,
1004535.08620228, 924307.9193291 , 964217.30193258,
1220872.72352639, 1266519.75438405, 1619407.66362994,
1582521.8990761 , 764723.64152428, 1739052.84077971,
1593274.88965288, 1677324.99055414, 982954.91497658,
1370603.38372502, 906582.85685703, 1631028.42785935,
690988.17277099, 1573202.0784274 , 1466498.88062096,
1650267.04845535, 1182468.07832065, 1173494.86869985,
1333497.9602431 , 1256991.46375876, 1859915.69316212,
758934.10499929, 1392542.74052307, 1711869.49819383,
1561607.5874123 , 886776.00191142, 1312020.95776119,
1174661.91515481, 1071955.83874449, 1381315.05056356,
1106592.45010921, 1170614.82399694, 992963.00044671,
1471298.27839551, 1531571.13792432, 1609332.54145859,
1374687.26168907, 963514.88833804, 1516653.63997645,
1161808.9337429 , 874674.36305494, 1168468.71316205,
608242.40551401, 1499895.89008379, 1020508.80251927,
1657914.86180067, 870372.4297565 , 1469286.71372917,

```

1585005.1099001 , 2474725.66527174, 1312108.9754553 ,
  927776.12992536, 1291387.28478204, 1591848.64102514,
  601180.27473501, 1595036.52624213, 1186556.58305162,
1569918.89962094, 775297.74281862, 1498585.74619838,
  731227.25771155, 1504692.54832438, 1046686.00738128,
1305402.58575973, 1275949.48895682, 997331.42657978,
  617836.40659196, 1591112.86332926, 1400763.16673385,
1311860.51933239, 1197044.05948832, 1041880.02840154,
1530068.19561792, 967934.24578182, 1427009.14149094,
1165147.25822145, 869045.49036599, 1541101.84334551,
  952854.45148333, 1649273.51933494, 1262902.39001229,
  630780.8541972 , 576868.98552437, 1709538.25511068,
1627857.63001048, 838666.12045459, 878776.97048247,
1299345.44781405, 1160416.6213845 , 749980.45207412,
  983974.19311116, 1153309.90526939, 943191.01927427,
1123570.59504184, 1547446.72609461, 1298891.03780601,
  694670.38901478, 1042411.7239971 , 858369.88200498,
1238168.96478328, 1069914.1618376 , 1032803.77343633,
1215816.40190439, 1344361.73315558, 1128683.07321975,
1451580.2340314 , 1094467.12736013, 1242819.75475191,
1312634.56732406, 1501591.70729071, 1577868.55728424,
1263870.34957635, 966363.73113454, 1304929.99505732,
  948730.96917438, 1116943.51964661, 1177297.49010983,
  988220.39094923, 1035263.71039961, 1114925.57767774,
1691529.13991383, 493943.88206347, 1819479.31451043,
1387271.64234034, 632696.71378614, 1224357.78570385,
1180743.31764195, 1036241.70518232, 921234.34851234,
1754054.1117938 , 1583328.06187218, 1563758.96394329,
1342866.59984841, 1173594.23690919, 1042902.5754118 ,
1318531.80156886, 1870001.48200049, 1663623.8581757 ,
1045096.55057783])

```

y_test

```

1501    1.339096e+06
2586    1.251794e+06
2653    1.340095e+06
1055    1.431508e+06
705     1.042374e+06

```

```

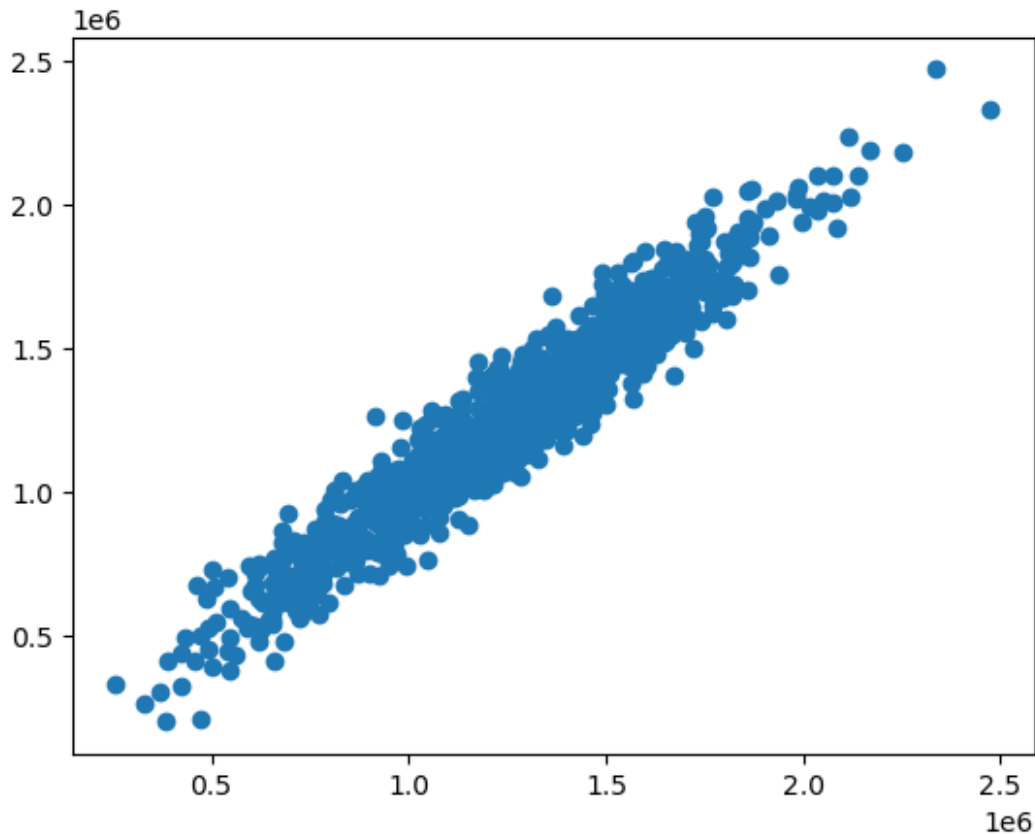
...
4711    1.107031e+06
2313    1.405505e+06
3214    1.924156e+06
2732    1.571254e+06
1926    8.831475e+05

```

Name: Price, Length: 1000, dtype: float64

```
plt.scatter(y_predict,y_test)
```

<matplotlib.collections.PathCollection at 0x28530443e90>



```
from sklearn.metrics import mean_squared_error
```

```
mse=mean_squared_error(y_test,y_predict)
```

```
mse
```

```
10089009300.890661
```

```
np.sqrt(mse)
```

```
100444.06055556824
```

```
rmse=np.sqrt(mse)
```

```
rmse
```

```
100444.06055556824
```

```
x_anish=[[80000,1,4,2,50000]]
```

```
model1.predict(x_anish)
```

```
C:\Users\srini\anaconda3\Lib\site-packages\sklearn\base.py:493:  
UserWarning: X does not have valid feature names, but LinearRegression  
was fitted with feature names  
warnings.warn(  

```

```
array([508662.51652624])
```

```
x_dimpy=[[100000,2,6,4,80000]]
```

```
model1.predict(x_dimpy)
```

```
C:\Users\srini\anaconda3\Lib\site-packages\sklearn\base.py:493:
```

```
UserWarning: X does not have valid feature names, but LinearRegression  
was fitted with feature names
```

```
warnings.warn(
```

```
array([1808611.29510401])
```